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LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

1.Problem Statement: Disaster Relief Supply & Volunteer Coordinator

The system helps disaster relief coordinators manage disaster requests, relief supplies, inventory, and volunteers during emergency situations.

REQUIREMENTS TABLE

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-01	Functional	Manage disaster requests	High	Create/update/ view requests	Tracks relief needs
FR-02	Functional	Allocate relief supplies	High	Supplies allocated	Ensures proper distribution
FR-03	Functional	Manage volunteers	High	Register/update volunteers	Supports coordination
FR-04	Functional	Assign volunteers	High	Assign available volunteers	Helps relief work
FR-05	Functional	Track inventory	High	Stock updated correctly	Avoids stock errors

FR-06	**Functional**	**Store system records**	**High**	**Data saved/retrieved**	**Centralized information**
NFR-01	**Security**	**Protect user data**	**High**	**Unauthorized access blocked**	**Improves security**
NFR-02	**Performance**	**Fast request processing**	**High**	**Quick response**	**Important in emergencies**
NFR-03	**Reliability**	**Reliable operation**	**High**	**Operations complete correctly**	**Ensures dependability**
NFR-04	**Maintainability**	**Easy to modify**	**Medium**	**Components change independently**	**Easy maintenance**
NFR-05	**Scalability**	**Handle more users/data**	**Medium**	**Supports increased load**	**Large emergencies**
NFR-06	**Usability**	**Simple interface**	**Medium**	**Easy common tasks**	**Reduces errors**

2. Architecture Selection

We selected Layered Architecture because the system has clearly separated responsibilities for user interaction, request processing, volunteer management, supply management, and data storage.

3. Component Identification

Based on the selected **Layered Architecture**, the Disaster Relief Supply & Volunteer Coordinator System is divided into the following components:

3.1 Coordinator / Volunteer UI

Provides the user interface through which coordinators and volunteers can submit requests, view information, check assignments, and monitor relief activities.

3.2 Disaster Request Manager

Manages disaster relief requests, including creating, updating, prioritizing, and tracking the status of requests.

3.3 Supply & Request Manager

Processes supply requirements and coordinates the allocation of available relief supplies to disaster requests.

3.4 Volunteer Management Service

Manages volunteer registration, availability, assignment, and volunteer status during relief operations.

3.5 Supply Inventory Manager

Maintains information about available relief supplies, quantities, stock updates, and inventory availability.

3.6 Database Manager

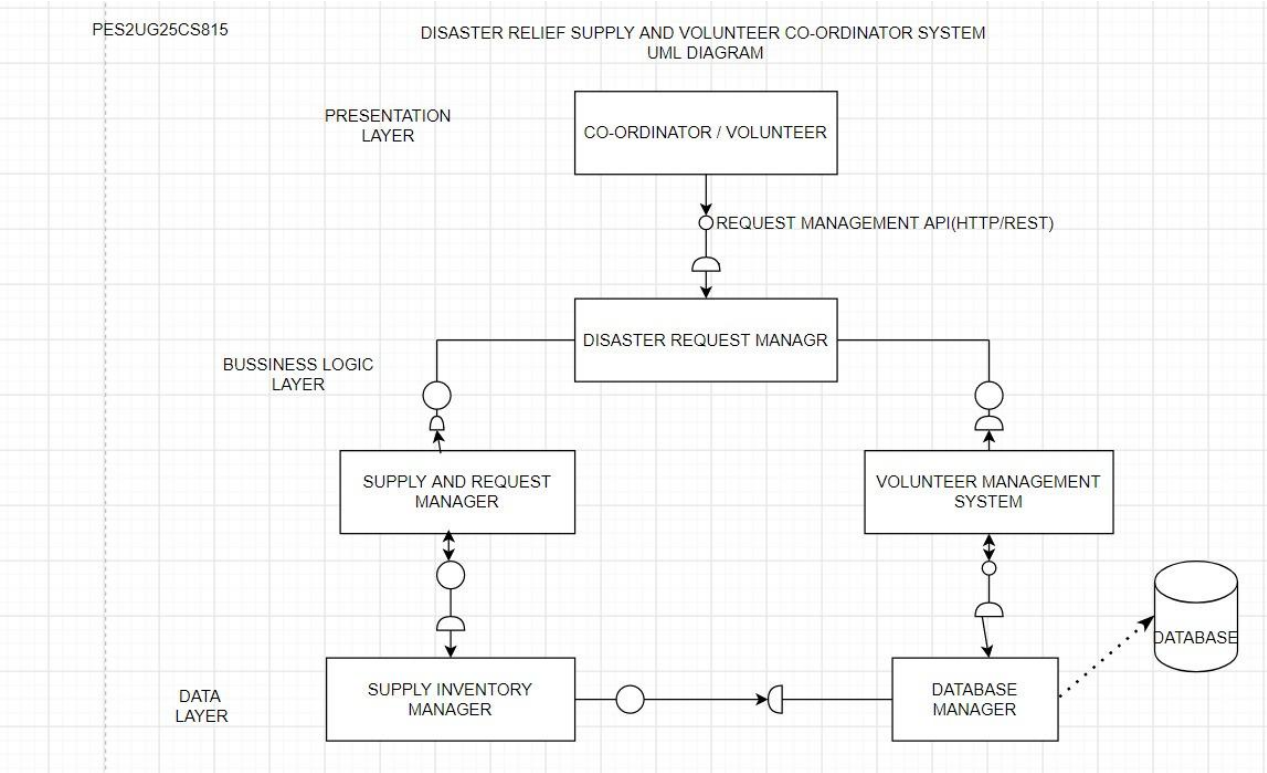
Stores and retrieves disaster requests, volunteer details, supply information, inventory records, and allocation details.

Interfaces

The following interfaces are used for communication between components:

1. **Request Management API** – Coordinator / Volunteer UI ↔ Disaster Request Manager
2. **Supply Allocation API** – Disaster Request Manager ↔ Supply & Request Manager
3. **Volunteer Assignment API** – Disaster Request Manager ↔ Volunteer Management Service
4. **Inventory API** – Supply & Request Manager ↔ Supply Inventory Manager
5. **Volunteer Data API** – Volunteer Management Service ↔ Database Manager
6. **Database Query Interface** – Supply Inventory Manager ↔ Database Manager

4. UML DIAGRAM



Architectural Choice

The system is divided into three logical layers: Presentation Layer, Business Logic Layer, and Data Layer. The Presentation Layer contains the Coordinator/Volunteer UI. The Business Logic Layer contains the Disaster Request Manager, Supply & Request Manager, and Volunteer Management Service. The Data Layer contains the Supply Inventory Manager and Database Manager.

Reason 1 – Separation of Responsibilities

Layered Architecture separates user interaction, business processing, and data management. For example, disaster requests can be processed by the Disaster Request Manager without directly accessing the database. This makes the system easier to understand and maintain.

Reason 2 – Easy Maintenance

The system may require changes such as adding new relief supplies, volunteer roles, or request-priority rules. With separate layers and components, these changes can be implemented with minimal impact on other components.

Security Advantage

The Presentation Layer does not directly access the database. Requests pass through business components where authentication, authorization, and input validation can be performed before accessing disaster, volunteer, and inventory information.

Performance Benefit

Business components communicate only with the components required for a particular operation. For example, the Supply & Request Manager can directly communicate with the Supply Inventory Manager to check supply availability, reducing unnecessary processing.

Conclusion

Layered Architecture is suitable for the Disaster Relief Supply & Volunteer Coordinator because it provides separation of responsibilities, easy maintenance, controlled data access, and efficient processing of disaster requests, supplies, and volunteer assignments.